# KARUNYA INSTITUTE OF TECHNOLOGY AND SCIENCES

School of Computer Science & Technology — Division of Computer Science & Engineering

# IA-3 CONTINUOUS INTERNAL EXAMINATION MASTER CHECKLIST

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|---|---|---|---|
| **Course Title & Code:** | **25CS208 - Web Technology** | **Academic Year:** | **2026–2027** |
| **Semester / Target:** | **Odd Semester (1st Year B.Tech CSE)** | **Assessment Type:** | **IA-3 Micro Project & E-Report** |

**STUDENT NAME(S):** _____

**REGISTER NO(S):** _____

**PROJECT TITLE:** _____

## 1. PROJECT GENERIC REQUIREMENTS & TECHNICAL VERIFICATION

☐ **FR-01: Landing Page (`index.html`)**

- Header navigation bar containing brand logo, section navigation links, and theme toggle button.
- Hero banner section featuring project title, subtitle introduction, and primary call-to-action buttons.
- Product feature overview cards and responsive document outline.

☐ **FR-02: Registration Page & Detailed Form Validation (`register.html`)**

- Form fields for Full Name, Employee / Register ID (`empId`), Email Address, Mobile Number, Division Dropdown (`dept`), Password, and Confirm Password.
- Real-time client-side Regex input validation executing on field `blur` and form `submit` events:
  - **Full Name:** Alphabets and spaces only (`/^[a-zA-Z\s]{2,50}$/`).
  - **Employee / Reg ID:** 6–12 alphanumeric characters (`/^[A-Z0-9]{6,12}$/i`).
  - **Email Address:** Valid format check (`/^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/`).
  - **Mobile Number:** 10-digit number starting with 6–9 (`/^[6-9]\d{9}$/`).
  - **Division Selection:** Mandatory selection check (`value !== ""`).
  - **Password Complexity:** Min 8 chars with uppercase, lowercase, number & symbol (`@$!%*?&`).
  - **Confirm Password:** Matching validation against password input value.
- Dynamic inline visual error badges (`<span class="error-msg">`) and red field highlights (`input.invalid`) on failed validation.
- Submit button disabled (`button.disabled = true`) until all Regex and selection checks evaluate to valid (`input.valid`).

☐ **FR-03: Sign-Up Page (`signup.html`)**

- Simplified account onboarding form with email/password creation and Terms of Service consent checkbox.

☐ **FR-04: Sign-In Page (`signin.html`)**

- Login credentials interface with password show/hide eye toggle button (`type="password"` ↔ `type="text"`).
- Dynamic client-side error alert modal/banner displayed on invalid credential format attempts.

☐ **FR-05: Home Page Dashboard & Dual Responsive Layouts (`home.html`)**

- **Desktop Layout (≥768px):** Multi-column dashboard structured via CSS Grid (`display: grid; grid-template-columns: 250px 1fr 300px;`).
- **Mobile Layout (<768px):** Single-column stacked stream via CSS Flexbox (`display: flex; flex-direction: column;`) with dynamic hamburger navigation drawer.

- ☐ **FR-06: Assistant Chatbot Integration (`chatbot.js`)**
  - Floating Action Button (FAB) anchored in the bottom-right viewport corner (`position: fixed`).
  - Expandable chat panel displaying dynamic user (`.user-msg`) and automated bot (`.bot-msg`) message nodes.
  - Keyword-matching reply engine with auto-scrolling log (`scrollTop = scrollHeight`).
- ☐ **FR-07: Light / Dark Mode Toggle Switcher (`theme.js`)**
  - Global theme toggle button integrated inside the primary navigation header bar.
  - CSS custom variable tokens (`:root` color definitions) switching light/dark color schemes instantly.
  - Persistent user theme selection saved across sessions using `window.localStorage`.
- ☐ **FR-08: JavaScript Calculator Module (`calculator.js`)**
  - Interactive key grid (digits 0–9, arithmetic operators `+`, `-`, `*`, `/`, Clear `C`, Equals `=`).
  - Digital output display screen with error handling preventing zero-division or invalid mathematical expressions.

## 2. CODE REPOSITORY & CLOUD DEPLOYMENT

- ☐ **Git & GitHub Version Control:** Codebase fully pushed to GitHub repository with clean commit history and proper directory organization.
- ☐ **Vercel Live Production Hosting:** Project deployed and publicly accessible via live Vercel URL (`https://project-name.vercel.app`) with active HTTPS SSL security.

## 3. MICRO PROJECT E-REPORT (BOUND PAPER & PDF SUBMISSION)

### Document Formatting Guidelines Compliance

- ☐ **Page Count:** Minimum 20 pages total.
- ☐ **Page Format:** A4 Portrait size, Normal Margins (1 inch / 2.54 cm).
- ☐ **Typography & Layout:** Times New Roman, 12 pt body font size, 1.5 line spacing, fully justified text alignment.

### Report Sequence & Required Chapters

- ☐ **Cover / Title Page:** Official template formatted with Project Title, Course Code (`25CS208 Web Technology`), Academic Year (`2026-27`), Student Names & Reg. Numbers, and Supervisor Name.
- ☐ **Bonafide Certificate:** Signed faculty supervisor certificate page.
- ☐ **Abstract:** 250–300 word executive summary detailing problem domain, tech stack, features, and Vercel hosting result.
- ☐ **Declaration & IPR Acknowledgment:** Official IPR agreement signed by student lead acknowledging the 60:40 commercial royalty policy.
- ☐ **Table of Contents:** Complete chapter, section, sub-section, and appendix index table.
- ☐ **Chapter 1: Introduction:** Problem background, industry use case, project objectives, and UN SDG / KITS technology mission link.
- ☐ **Chapter 2: Literature Review & Tech Stack:** Survey of existing web portals, technology stack justification matrix, and student team role allocation table.
- ☐ **Chapter 3: System Requirements & SRS:** IEEE 830 specification, scope boundaries (in-scope vs. out-of-scope), functional requirement breakdown (FR-01 to FR-08), and non-functional standards.
- ☐ **Chapter 4: Design & Code Implementation:** Figma wireframe diagrams, directory folder tree, desktop CSS Grid vs. mobile CSS Flexbox implementation, validation logic, theme switcher, JS calculator, and chatbot engine.
- ☐ **Chapter 5: Testing, Results and Deployment:** Form Regex input test matrix, cross-device viewport test results, GitHub commit documentation, and Vercel production hosting proof.

- ☐ **Chapter 6: Conclusion and Future Scope:** Key technical learnings recap and Version 2.0 feature roadmap.
- ☐ **References:** Standard IEEE formatted references for W3C, MDN, IEEE standards, and Web Tech sources.
- ☐ **Appendix A: Formatted & Optimised Source Code (TEXT ONLY — NO CODE SCREENSHOTS):**
  - A.1 Formatted HTML5 text code (`index.html`, `register.html`, `signup.html`, `signin.html`, `home.html`).
  - A.2 Formatted CSS layout text code (`styles.css`, `layout.css`).
  - A.3 Formatted JavaScript text code (`validation.js`, `theme.js`, `calculator.js`, `chatbot.js`).
- ☐ **Appendix B: Application Screenshots Gallery & Traceability:**
  - B.1 Functional Requirement (FR) to Screenshot Mapping Matrix table.
  - B.2 High-resolution screenshots of Landing, Registration, Sign-Up, Sign-In, and Home pages.
  - B.3 Side-by-side screenshots comparing Desktop CSS Grid vs. Mobile CSS Flexbox viewports.
  - B.4 Screenshots of active Light/Dark theme states and JS Calculator operations.
  - B.5 Screenshots of form Regex validation alerts and active Chatbot conversation window.
  - B.6 Screenshots of GitHub repository commits and live Vercel HTTPS dashboard.

## 4. FORM SUBMISSION & HARD COPY PRINTOUTS

- ☐ **Google Form Submission:** Uploaded digital PDF report, repository link, Vercel link, and student metadata.
- ☐ **Hard Copy Printouts (Physical Documents):**
  - Project Proposal printout.
  - Official Assessment Rubric sheet.
  - Micro Project E-Report (20+ pages bound hard copy).

## 5. YOUTUBE PRESENTATION VIDEO (5-MINUTE LIMIT)

- ☐ **1. Introduction:** Introduce yourself and all team members.
- ☐ **2. Course Details:** State Course Name (`Web Technology`) and Course Code (`25CS208`).
- ☐ **3. Project Title:** State the exact title of your project.
- ☐ **4. Problem Statement:** Explain the problem your application addresses.
- ☐ **5. Technologies Used:** Detail the tech stack (HTML5, CSS3 Grid/Flexbox, ES6+ JS, Git, Vercel).
- ☐ **6. Solution:** Overview of system architecture and core portal features.
- ☐ **7. Roles:** Outline team member roles and individual technical contributions.
- ☐ **8. Website Walkthrough:** Demonstrate the live website on both desktop (CSS Grid) and mobile (CSS Flexbox) viewports.
- ☐ **9. Validation:** Demonstrate real-time form validation by triggering invalid Regex error states live.
- ☐ **10. Code Optimization:** Highlight modular code structure, CSS design tokens, and JS event handlers.
- ☐ **11. Live Hosting & QR Code:** Show the live `.vercel.app` website link and display the **Live Website QR Code** on screen.
- ☐ **12. Repository QR Code:** Display the **Git Repository QR Code** on screen.